

MITRE ATLAS in Telecom AI

Mapping the Adversarial Threat Landscape for AI Systems to Telecom AI Attack Surfaces

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Executive Summary

MITRE ATLAS (Adversarial Threat Landscape for Artificial-Intelligence Systems) is the closest thing the security community has to an ATT&CK-style framework for AI/ML systems. It catalogs 16 tactics, well over 100 techniques and sub-techniques, roughly 30+ mitigations, and 57 real-world case studies (as of the current release cycle). It is model-agnostic and industry-agnostic: it was not written for telecom, and it currently contains no telecom-specific tactic, technique, or case study.

Telecom operators, however, are now deploying AI/ML across nearly every layer of the network: RF signal classification and spectrum sensing, O-RAN near-real-time RIC xApps that make live radio-resource decisions, 5G Core analytics (NWDAF), autonomous RAN optimization (e.g. Nokia's MantaRay AutoPilot, Ericsson's multi-agent configuration tools), network-operations LLM copilots (Nokia + Azure AI Foundry, Ericsson + Mistral AI), and customer-facing chatbots and fraud/identity models. Each of these is a plausible target for the same adversary behaviors ATLAS catalogs elsewhere.

This document does two things: (1) summarizes the ATLAS framework itself, and (2) walks through all 16 ATLAS tactics and maps each to a concrete telecom AI surface, an illustrative attack scenario grounded in current research and known incident patterns, and a mitigation direction. Where a technique maps closely to an existing academic finding (e.g. O-RAN xApp KPI poisoning) or an existing ATLAS case study pattern (e.g. deepfake bypass of identity verification), that lineage is called out.

Scoping note: the tactic/technique names and IDs below are drawn directly from the official ATLAS knowledge base (atlas.mitre.org). The telecom scenarios are this document's original extension of that framework, built from current O-RAN/5G security research and public 2025-2026 telecom AI deployments. They are not text copied from ATLAS itself, since ATLAS does not yet publish telecom case studies.

1. What MITRE ATLAS Is

ATLAS is a living, community-maintained knowledge base of adversary tactics and techniques against AI-enabled systems, built and hosted by MITRE and modeled

directly on the structure of MITRE ATT&CK. It exists because AI components (training data, model weights, inference APIs, prompts, RAG pipelines, agent tool integrations) introduce attack surfaces that classic IT-focused frameworks don't describe well.

Framework anatomy:

- **Tactics:** 16 top-level adversary goals (“why”), from Reconnaissance through Impact.
- **Techniques and sub-techniques:** over 100 specific methods (“how”) an adversary uses to accomplish a tactic; several are “dual-mapped” to more than one tactic (e.g. LLM Jailbreak sits under both Privilege Escalation and Defense Evasion).
- **Mitigations:** defensive controls mapped to the techniques they counter; roughly 70% overlap with conventional security controls already used in SOC workflows, the rest AI-specific (e.g. adversarial input detection, model provenance).
- **Case studies:** 57 documented real-world incidents and red-team exercises, spanning facial recognition evasion, chatbot data poisoning, model theft, prompt-injection RCE, and AI-agent supply-chain compromise.

ATLAS is complementary to, not a replacement for, ATT&CK: an attack on an AI-enabled telecom system typically starts and ends in conventional IT/OT territory (phishing, credential theft, network pivoting), with an ATLAS-specific stage in the middle, such as model poisoning, adversarial evasion, or prompt injection.

2. Where AI Actually Sits in Telecom Networks

Before mapping threats, it helps to be concrete about where ML/AI is actually deployed in a modern telecom stack, since that is what defines the attack surface:

- **Physical / RF layer:** deep-learning signal classification, spectrum sensing, and modulation recognition.
- **O-RAN near-real-time RIC:** ML-based xApps making live decisions on handover, traffic steering, and radio resource management (RRM); O-RAN Alliance WG11 has published dedicated threat models for this layer.
- **5G Core:** NWDAF (Network Data Analytics Function) consuming network and subscriber data to drive automated policy decisions.
- **Autonomous network operations:** AI-RAN and self-optimizing-network tools such as Nokia's MantaRay AutoPilot (deployed with NTT DOCOMO) and Ericsson's multi-agent configuration/troubleshooting tools, which can automatically design network parameters and execute real-time adjustments.

- **Network-ops copilots:** LLM-based assistants with tool-calling access to OSS/BSS systems, e.g. Nokia's copilot built on Azure AI Foundry, Ericsson's collaboration with Mistral AI for agent co-development.
- **Customer-facing AI:** support chatbots, churn prediction, and fraud/identity models (SIM-swap risk scoring, KYC verification, voice biometrics).

Every one of these is, at bottom, an AI model or AI-integrated agent with training data, an inference interface, and, increasingly, autonomous tool-calling privileges. That is exactly the surface ATLAS was built to describe.

3. Tactic-by-Tactic Mapping to Telecom AI

The table below walks through all 16 ATLAS tactics in their canonical kill-chain order and maps each to a telecom-specific manifestation.

ATLAS Tactic	Example Technique(s)	Telecom AI Surface	Illustrative Scenario	Mitigation Direction
1. Reconnaissance (AML.TA0002)	Active Scanning (T0006); Gather RAG-Indexed Targets (T0064)	Network-ops copilots & RAG knowledge assistants (e.g. Nokia/Azure copilot, Ericsson agent tools)	Attacker scrapes public API docs and RAG-indexed runbooks to map an operator's O-RAN/NWDAF architecture before an attack.	Restrict public exposure of internal API docs; rate-limit and monitor scraping of copilot-facing endpoints.
2. Resource Development (AML.TA0003)	Obtain Capabilities (T0016); Publish Poisoned Models (T0058)	Open-source xApp/RIC ML models; pretrained models pulled into fraud or RF classifiers	Adversary publishes a poisoned RF signal-classification model to a public model hub that an operator's MLOps pipeline later ingests.	Model provenance verification, signed artifacts, AI-specific SBOM and vetting before pipeline ingestion.
3. Initial Access (AML.TA0004)	AI Supply Chain Compromise (T0010); Exploit Public-Facing Application (T0049)	5G Network Exposure Function (NEF) / NWDAF APIs; RIC xApp container images	Attacker exploits an exposed NEF API feeding NWDAF analytics, or compromises an xApp container image in the CI/CD chain.	API gateway hardening and authentication on NEF/NWDAF interfaces; container image scanning; zero-trust onboarding for xApps.
4. AI Model Access (AML.TA0000)	AI Model Inference API Access (T0040); AI-Enabled Product or Service (T0047)	Customer-facing chatbots; churn/fraud-scoring inference endpoints	Attacker obtains black-box query access to a churn or fraud-scoring model through the customer mobile app's API.	Rate limiting, query auditing, and output perturbation on customer-facing inference APIs.
5. Execution (AML.TA0005)	LLM Prompt Injection (T0051); AI Agent Tool Invocation (T0053)	Network-ops LLM copilots with tool-calling access to OSS/BSS systems	Indirect prompt injection hidden in a trouble-ticket description causes the copilot agent to invoke a "reconfigure cell" tool.	Tool-invocation allowlists, human approval for state-changing actions, prompt-injection filtering on ingested text.

ATLAS Tactic	Example Technique(s)	Telecom AI Surface	Illustrative Scenario	Mitigation Direction
6. Persistence (AML.TA0006)	Poison AI Model (T0018.000); RAG Poisoning (T0070)	RAG corpora backing network knowledge-base copilots	Adversary persistently poisons ingested runbook documents so the copilot repeatedly recommends an insecure configuration.	Integrity checks and version control on RAG source documents; audited, permissioned knowledge-base updates.
7. Privilege Escalation (AML.TA0012)	LLM Jailbreak (T0054); AI Agent Tool Invocation (T0053)	Ops copilots holding elevated OSS/BSS permissions	A jailbreak prompt escalates a copilot from read-only diagnostics to executing privileged network-configuration commands.	Least-privilege tool scopes; separate read-only and write-capable agent roles; jailbreak-resistant system prompts.
8. Defense Evasion (AML.TA0007)	Evade AI Model (T0015); Impersonation (T0073)	RF/spectrum-sensing classifiers; voice-biometric IVR authentication	GAN-crafted RF waveform evades a spectrum-sensing classifier; a deepfake voice impersonates a subscriber to pass voice authentication for a SIM swap.	Adversarial-robust training for signal classifiers; liveness detection; multi-factor identity verification beyond voice alone.
9. Credential Access (AML.TA0013)	Unsecured Credentials (T0055)	MLOps pipelines, model registries, xApp configuration files	Hardcoded NEF or RIC API keys are found embedded in a poorly secured xApp container configuration.	Centralized secrets management; automated credential scanning in CI/CD pipelines.
10. Discovery (AML.TA0008)	Discover AI Artifacts (T0007); Discover LLM System Information (T0069)	Customer-service LLM system prompts; RIC model registries	Prompt-probing extracts a customer-service LLM's system instructions, revealing internal API structure and escalation paths.	System-prompt hardening, output filtering, and canary tokens to detect extraction attempts.
11. Lateral Movement (AML.TA0015)	Phishing (T0052)	NOC and customer-support staff workflows	AI-generated deepfake voice impersonates an executive to trick a NOC engineer into granting RIC/xApp deployment access.	Callback verification procedures for privileged requests; phishing-resistant MFA for NOC and support staff.
12. Collection (AML.TA0009)	Data from Information Repositories (T0036)	Subscriber data lakes feeding fraud, churn, and personalization models	Compromised MLOps access is used to collect training data containing subscriber PII beyond what the model actually needs.	Strict data access controls and PII minimization in training datasets; data-use audit logging.
13. AI Attack Staging (AML.TA0001)	Craft Adversarial Data (T0043); Create Proxy AI Model (T0005)	RRM/handover optimization models; RF signal classifiers in the near-RT RIC	Adversary trains a proxy model that mimics an operator's handover-optimization model to craft perturbations transferable to the real model.	Model ensembling and adversarial training; restrict inference-API access that would let an outsider build an accurate proxy.
14. Command	Reverse Shell	Compromised AI	A compromised NOC	Network

ATLAS Tactic	Example Technique(s)	Telecom AI Surface	Illustrative Scenario	Mitigation Direction
& Control (AML.TA0014)	(T0072)	agents with tool access to network infrastructure	automation agent's tool integration is used to establish a reverse shell into the management network.	segmentation of AI-agent execution environments; egress monitoring and anomaly detection.
15. Exfiltration (AML.TA0010)	Exfiltration via AI Inference API (T0024); LLM Data Leakage (T0057)	Support LLMs with CRM/subscriber-record access	Prompt injection causes a support LLM to leak subscriber PII or internal network topology details in its response text.	DLP on LLM outputs; strict segregation between the model's working context and sensitive backend systems.
16. Impact (AML.TA0011)	Denial of AI Service (T0029); Erode AI Model Integrity (T0031); External Harms (T0048)	RAN traffic-steering models; fraud-detection stack; customer-facing AI	Chaff-data flooding degrades a real-time RAN traffic-steering model, causing service degradation; eroded fraud-model integrity enables large-scale SIM-swap fraud.	Model health/drift monitoring; automatic fallback to rule-based safe modes; incident-response playbooks mapped to ATLAS mitigations.

4. Four Scenarios Worth Walking Through

4.1 O-RAN xApp KPI / model poisoning

Academic work on O-RAN near-RT RIC (e.g. “KPI Poisoning: An Attack in Open RAN Near Real-Time Control Loop,” and O-RAN Alliance WG11's threat model) documents that a malicious or compromised xApp with read/write access to the Shared Data Layer (SDL) or E2 interface can poison the KPIs and telemetry that other xApps' ML models train and infer on, biasing handover or traffic-steering decisions without ever touching the model file itself. In ATLAS terms this spans Resource Development (poisoned model/data), Initial Access (supply-chain compromise of the xApp), Persistence (Poison AI Model), and ultimately Impact (Erode AI Model Integrity).

4.2 Prompt injection against a network-ops copilot

As operators like Nokia and Ericsson roll out LLM-based copilots with direct tool-calling access to configuration and ticketing systems, indirect prompt injection becomes a realistic path from a low-privilege input (a customer complaint, a trouble ticket, a scraped web page) to a privileged action (Execution via AI Agent Tool Invocation, potentially escalating via LLM Jailbreak). This mirrors ATLAS case studies such as the Jira Service Management and Copilot Studio indirect-injection incidents, extended to a telecom OSS/BSS context.

4.3 Deepfake-assisted SIM-swap / identity fraud

Telecom is a high-value target for voice-deepfake and synthetic-identity fraud because SIM swaps and account recovery routinely rely on voice or knowledge-based verification. ATLAS's Impersonation (T0073) technique and its deepfake-related case studies (e.g. live deepfake injection to evade mobile KYC verification) map directly onto this: an adversary uses a cloned voice or synthetic video to defeat an operator's identity-verification model, then pivots that access into account takeover or number-porting fraud.

4.4 Adversarial evasion of RF/spectrum classifiers

Research on adversarial ML for 5G/6G shows that GAN-crafted or optimally perturbed RF waveforms can cause deep-learning-based signal classifiers and physical-layer authentication systems to misclassify traffic or spoof authorized transmitters. This is a textbook case of ATLAS's Evade AI Model (T0015) and Craft Adversarial Data (T0043), staged via a proxy model trained on eavesdropped signal data (Create Proxy AI Model, T0005).

5. Mitigation Themes

ATLAS mitigations translate into a small number of recurring controls that cover most of the table above:

Control Area	Applies To
AI/ML supply-chain provenance (signed models, SBOM, registry vetting)	Resource Development, Initial Access, Persistence
Least-privilege, human-approved tool invocation for agents/copilots	Execution, Privilege Escalation, Command & Control
Adversarial-robust training and input validation	Defense Evasion, AI Attack Staging
RAG/knowledge-base integrity and versioning	Persistence, Discovery
Output filtering and DLP on model/agent responses	Exfiltration, Discovery
Model health, drift, and integrity monitoring with rule-based fallback	Impact
Phishing-resistant MFA and callback verification for privileged requests	Credential Access, Lateral Movement

6. Sources

- **MITRE ATLAS:** official knowledge base, atlas.mitre.org
- **MITRE ATLAS data repository:** tactics, techniques, case studies: github.com/mitre-atlas/atlas-data
- **O-RAN Alliance WG11 threat modeling:** “KPI Poisoning: An Attack in Open RAN Near Real-Time Control Loop” (arXiv:2505.05537)
- **Near-RT RIC ML security research:** “Exploiting and Securing ML Solutions in Near-RT RIC” (arXiv:2406.12299); “Experimental Study of Adversarial Attacks on ML-based xApps in O-RAN” (arXiv:2309.03844)
- **5G/6G adversarial ML research:** “Adversarial Machine Learning for 5G Communications Security” (arXiv:2101.02656); “Security and Privacy in O-RAN for 6G” (arXiv:2603.21211)
- **Nokia network-ops copilot:** Nokia / Microsoft Azure AI Foundry; Nokia MantaRay AutoPilot with NTT DOCOMO, Microsoft Cloud Blog, February 2026
- **Ericsson AI agent tools:** Ericsson-Mistral AI partnership, Ericsson white papers and news releases, 2025-2026